

Evaluation Report for Riya Shyam Huddar - Assignment 1

Notebook Link: https://colab.research.google.com/drive/1f_bobightTAsD64sRQP18RDviwmFY7fX

Evaluation Summary

The submission correctly implements and analyzes statistical laws for three language corpora: English, Marathi, and Hindi. Preprocessing is adequately performed, and Zipf's Law is correctly fitted with both Zipf and Mandelbrot models, with the latter consistently showing a better fit. Heaps' Law is modeled using both non-linear curve fitting and linear regression, and the parameters are compared. The computational scaling analysis is accurate, correctly concluding the infeasibility of storing a full vocabulary-document matrix. The final report effectively summarizes and compares the findings across languages, though the repeated execution of plotting and calculation code in the final summary section is redundant.

Metric	Score
Obtained	105.00 / 105.00
Percentage	100.0%

Rubric Reference

Category	Deliverable/Criterion	Max Marks	Automatic Scoring Logic (Objective Check)
I. General & Submission (5 Marks)			
G1. Naming Convention	Filename adheres to First Name + Roll Number + _Assignment01.ipynb (e.g., SriramBMC202204_Assignment01.ipynb).	0 (Deduction only)	Deduct 0 marks if the notebook's file name does not strictly match the required format.
G2. Code Execution	Notebook runs from top-to-bottom without critical errors (e.g., cell crashes, import errors, file read failures).	5 (Bonus)	Award 5 marks if the notebook executes fully. Deduct 5 marks if execution fails early or major parts of the assignment cannot be run.
Task 1: Zipf's Law Analysis (20 Marks)			

T1.1. Log-Log Plots	3 plots (one for each language) showing log(rank) vs. log(frequency).	6 (2 per plot)	Award 2 marks per language if a plot object/image file named or captioned as "Zipf's Law Log-Log Plot" is detected, and it contains data points and lines for both the Zipf and Mandelbrot models.
T1.2. Parameter Reporting	Table reporting α (Zipf) and β (Mandelbrot) parameters for 3 languages.	6 (2 per language)	Award 2 marks per language if the output/table clearly reports a numerical value for α and β . (Check if values fall within an acceptable pre-determined range for sanity.)
T1.3. Fit Metrics Reporting	Table reporting R^2 and/or RMSE for both models for 3 languages.	6 (2 per language)	Award 2 marks per language if the output/table clearly reports at least one quantitative metric (R^2 or RMSE) for both Zipf and Mandelbrot fits.
T1.4. Top 5% Words	List of the top 5% most frequent words is generated and displayed for all 3 languages.	2	Award 2 marks if the output cell for each language displays a list of words corresponding to the top 5% of tokens.
Task 2: Heaps' Law - Vocabulary Growth (20 Marks)			
T2.1. Vocabulary Growth Plots	3 plots (one for each language) showing $V(N)$ vs. N (Vocabulary Size vs. Corpus Size).	9 (3 per plot)	Award 3 marks per language if a plot object/image file named or captioned as "Heaps' Law Vocabulary Growth" is detected, and it displays the empirical data points and the fitted Heaps' law curve.
T2.2. Parameter Reporting	Report/Table of estimated parameters K and β (from direct curve fitting) for 3 languages.	9 (3 per language)	Award 3 marks per language if the output/table clearly reports a numerical value for K and β derived from the $V(N)=K \cdot N^{\beta}$ fit.

T2.3. Successful Data Generation	The code successfully generates the $V(N)$ vs. N data points across the corpus size.	2	Award 2 marks if the underlying arrays for $V(N)$ and N are present and non-empty.
Task 3: Computational Scaling (20 Marks)			
T3.1. Memory Calculation	Correct calculation of memory requirement based on the formula: $\text{Memory} = (\text{Vocabulary Size}) \times (\text{Number of Documents}) \times 8 \text{ bytes}$	15 (5 per language)	Award 5 marks per language if the reported memory estimate (in GB or TB) is calculated using the corpus's reported Vocabulary Size and Number of Documents (wiki pages). (Check if the final reported value is numerically consistent with the input metrics.)
T3.2. Feasibility Statement	A concluding statement regarding the machine's ability to handle the corpus size.	5	Award 5 marks if the summary table or an adjacent text cell contains a clear, non-empty, yes/no statement addressing whether the corpus is feasible for a typical machine, based on the calculation.
Task 4: Regression Fitting of Heaps' Law (20 Marks)			
T4.1. Regression Code Presence	Visible, runnable code block utilizing a machine learning library (e.g., or) for linear regression on log-transformed data.	5	Award 5 marks if a code cell imports and uses a regression model on log-transformed $V(N)$ and N data.
T4.2. Regression Parameter Reporting	Report/Table of K and β parameters derived specifically from the regression model for 3 languages.	5	Award 5 marks if the output/table reports the K and β parameters obtained after running the regression model.
T4.3. Evaluation Metrics	Reporting of R^2 or MSE for the regression fit for 3 languages.	5	Award 5 marks if the output/table reports at least one metric (R^2 or MSE) evaluating the regression model's fit.

T4.4. Regression Plots	3 plots (one for each language) showing the log-transformed data and the fitted regression line.	5	Award 5 marks if a plot object/image file is detected that displays $\log(V(N))$ vs. $\log(N)$ and the calculated linear regression line.
Task 5: Comparative Analysis and Report (20 Marks)			
T5.1. Zipf's Summary Table	A table comparing α values and identifying the best-fit model (Zipf/Mandelbrot) across all 3 languages.	5	Award 5 marks if a single table compares the α value and explicitly names the preferred model (based on metrics from T1.3) for each of the 3 languages.
T5.2. Heaps' Summary Table	A table comparing K and β values across all 3 languages.	5	Award 5 marks if a single table compares K and β (using values from either T2 or T4) for each of the 3 languages.
T5.3. Memory Summary Table	A table comparing the calculated Memory Requirements across all 3 languages.	5	Award 5 marks if a single table presents the final calculated memory size (GB/TB) for each of the 3 languages.
T5.4. Discussion Formatting	Discussions/interpretations are formatted using Markdown within the notebook (not plain text).	5	Award 5 marks if at least three distinct interpretation/discussion sections (T1, T4, T5) are present and formatted using non-plain-text Markdown features (e.g., headers, bullet points, bold text).

Detailed Rubric Breakdown

Item 1 (Cell 1)

Rubric Criteria: G2. Code Execution

Score: 5.0/5.0 - Full

Justification: Output: / --- Hindi --- / Regression-fitted K: 8.62, beta: 0.63. The notebook executed from top to bottom without any critical errors, as evidenced by the successful output of the final code cells.

Code Feedback:

Concept Explanation & Output Analysis: The notebook successfully runs all code cells, from data loading and preprocessing to analysis and plotting for all three languages, indicating a robust and error-free implementation.

Item 2 (Cell 30)

Rubric Criteria: T1.1. Log-Log Plots

Score: 6.0/6.0 - Full

Justification: Output: / Top 50 words: ('the', 'of', 'in', 'and', 'a', 'to', 'was', 'for', 'on', 'as', 'is', 'by', 's', 'with', 'from', 'at', 'he', 'that', 'his', 'it', 'an', 'new', 'were', 'also', 'first', 'which', 'are', 'be', 'had', 'this', 'one', 'or', 'after', 'not', 'their', 'has', 'they', 'its', 'but', 'during', 'who', 'american', 'redirect', 'patriots', 'have', 'align', 'two', 'her', 'center', 'time'). Log-log plots showing rank vs. frequency are correctly generated for all three languages (English in cell 30, Marathi in 41, Hindi in 53), including fits for both Zipf and Mandelbrot models.

Code Feedback:

Concept Explanation & Output Analysis: The student created a function to generate log-log plots for word rank versus frequency. Each plot correctly displays the empirical data along with the fitted curves for both Zipf's and Mandelbrot's laws, fulfilling the requirements for all three corpora.

Item 3 (Cell 31)

Rubric Criteria: T1.2. Parameter Reporting

Score: 6.0/6.0 - Full

Justification: Output: Zipf and Mandelbrot Parameters for English: / / Zipf Mandelbrot. Parameter tables in cells 31 (English), 42 (Marathi), and 54 (Hindi) clearly report the numerical values for (for both models) and (for Mandelbrot), as required.

Code Feedback:

Concept Explanation & Output Analysis: For each language, a clear and well-formatted pandas DataFrame is used to present the estimated parameters (α , C , β) for both the Zipf and Mandelbrot models. This presentation is effective and meets the rubric requirements.

Item 4 (Cell 31)

Rubric Criteria: T1.3. Fit Metrics Reporting

Score: 6.0/6.0 - Full

Justification: Output: Zipf and Mandelbrot Parameters for English: / / Zipf Mandelbrot. Parameter tables in cells 31 (English), 42 (Marathi), and 54 (Hindi) report quantitative fit metrics (R^2 and RMSE) for both Zipf and Mandelbrot models across all three languages.

Code Feedback:

Concept Explanation & Output Analysis: The analysis includes key goodness-of-fit metrics, R^2 and RMSE, for both models. These are presented in the same summary tables as the parameters, allowing for direct comparison of model performance for each language.

Item 5 (Cell 30)

Rubric Criteria: T1.4. Top 5% Words

Score: 2.0/2.0 - Full

Justification: Output: / Top 50 words: ('the', 'of', 'in', 'and', 'a', 'to', 'was', 'for', 'on', 'as', 'is', 'by', 's', 'with', 'from', 'at', 'he', 'that', 'his', 'it', 'an', 'new', 'were', 'also', 'first', 'which', 'are', 'be', 'had', 'this', 'one', 'or', 'after', 'not', 'their', 'has', 'they', 'its', 'but', 'during', 'who', 'american', 'redirect', 'patriots', 'have', 'align', 'two', 'her', 'center', 'time'). The code generates the list of the top 5% most frequent words for each language, and a sample of these words is displayed in the output of cells 30, 41, and 53.

Code Feedback:

Concept Explanation & Output Analysis: The implementation correctly identifies and stores the top 5% of most frequent words for each corpus. A sample (top 50) is printed to the output, and the full list is used in subsequent analysis cells to examine stopword distribution.

Item 6 (Cell 65)

Rubric Criteria: T2.1. Vocabulary Growth Plots

Score: 9.0/9.0 - Full

Justification: Output: / Estimated parameters for English: $K = 12.28$, $\alpha = 0.63$ / Fit metrics: $R^2 = 0.998$, $MSE = 210865805.72$. Plots displaying vocabulary size ($V(N)$) versus corpus size (N) are correctly generated for all three languages (English in cell 65, Marathi in 67, Hindi in 69). Each plot includes both the empirical data and the fitted Heaps' Law curve.

Code Feedback:

Concept Explanation & Output Analysis: The student correctly plots the empirical vocabulary growth against the number of tokens processed. The plots also include the curve derived from the fitted Heaps' Law parameters, providing a clear visual comparison of the model and the data for all three languages.

Item 7 (Cell 65)

Rubric Criteria: T2.2. Parameter Reporting

Score: 9.0/9.0 - Full

Justification: Output: / Estimated parameters for English: $K = 12.28$, $\alpha = 0.63$ / Fit metrics: $R^2 = 0.998$, $MSE = 210865805.72$. The output of cells 65, 67, and 69 clearly reports the estimated parameters K and α from the direct curve fitting of Heaps' Law for each of the three languages.

Code Feedback:

Concept Explanation & Output Analysis: The parameters K and α , derived from non-linear curve fitting, are explicitly printed below each vocabulary growth plot. This provides a direct and accessible report of the key Heaps' Law constants for each corpus.

Item 8 (Cell 65)

Rubric Criteria: T2.3. Successful Data Generation

Score: 2.0/2.0 - Full

Justification: Output: / Estimated parameters for English: $K = 12.28$, $\beta = 0.63$ / Fit metrics: $R^2 = 0.998$, $MSE = 210865805.72$. The function in cell 62 generates the $V(N)$ vs. N data. The successful generation of plots and fitted parameters in subsequent cells (e.g., cell 65) confirms that the underlying data arrays ('tokens_seen', 'vocab_sizes') are present and non-empty.

Code Feedback:

Concept Explanation & Output Analysis: A function is implemented to iterate through the corpus and record vocabulary size at fixed token intervals. The successful execution of this function and its use in subsequent fitting and plotting steps demonstrate that the required data points were generated correctly.

Item 9 (Cell 79)

Rubric Criteria: T3.1. Memory Calculation

Score: 15.0/15.0 - Full

Justification: Output: . The table generated in cell 79 shows memory estimates (e.g., 3.80 TB for English) that are numerically consistent with the formula (Vocabulary Size * Number of Documents * 8 bytes). The vocabulary sizes are taken from Task 2 and document counts from cell 76, and the calculations are correct for all three languages.

Code Feedback:

Concept Explanation & Output Analysis: The student correctly calculates the memory required for a dense vocabulary-document matrix. The implementation uses the previously determined vocabulary sizes and document counts, applies the correct formula, and formats the output into human-readable units (TB), demonstrating a correct application of the concept.

Item 10 (Cell 79)

Rubric Criteria: T3.2. Feasibility Statement

Score: 5.0/5.0 - Full

Justification: Output: . The table generated in cell 79 includes a column titled 'Feasible on Local Machine' which contains a clear 'No' statement for each language. This is further supported by a markdown statement in cell 80.

Code Feedback:

Concept Explanation & Output Analysis: The summary table explicitly includes a column to address feasibility, which is populated with a clear 'No'. An additional markdown cell reinforces this conclusion, satisfying the rubric requirement completely.

Item 11 (Cell 84)

Rubric Criteria: T4.1. Regression Code Presence

Score: 5.0/5.0 - Full

Justification: Output: / --- English --- / Regression-fitted K : 9.55, β : 0.64. Cell 83 defines a function 'heaps_regression' that uses 'sklearn.linear_model.LinearRegression' to perform a fit on log-transformed data. This function is successfully called in cell 84, proving the presence of runnable regression code.

Code Feedback:

Concept Explanation & Output Analysis: The code correctly implements the linearization of Heaps' Law by taking the logarithm of both variables. It then uses a standard machine learning library (scikit-learn) to perform linear regression, demonstrating the correct procedure for this task.

Item 12 (Cell 84)

Rubric Criteria: T4.2. Regression Parameter Reporting

Score: 5.0/5.0 - Full

Justification: Output: / --- English --- / Regression-fitted K: 9.55, beta: 0.64. The outputs of cells 84 (English), 85 (Marathi), and 86 (Hindi) explicitly report the K and parameters derived from the linear regression model, including the necessary exponentiation to recover K.

Code Feedback:

Concept Explanation & Output Analysis: The parameters K and obtained from the regression are clearly printed for each language. The implementation correctly identifies the slope as and the anti-log of the intercept as K.

Item 13 (Cell 84)

Rubric Criteria: T4.3. Evaluation Metrics

Score: 5.0/5.0 - Full

Justification: Output: / --- English --- / Regression-fitted K: 9.55, beta: 0.64 / R^2 (log-scale): 0.999. The outputs of cells 84, 85, and 86 report evaluation metrics (R^2 and MSE) for the regression fit on both the log-transformed and original scales for all three languages.

Code Feedback:

Concept Explanation & Output Analysis: For each language, the submission reports both R^2 and MSE for the regression model. It correctly calculates these metrics on the log-scale where the fit was performed and also on the original scale for comparison, showing a thorough evaluation.

Item 14 (Cell 84)

Rubric Criteria: T4.4. Regression Plots

Score: 5.0/5.0 - Full

Justification: Output: / --- English --- / Regression-fitted K: 9.55, beta: 0.64. Plots showing the log-transformed data ($\log(V(N))$ vs. $\log(N)$) and the fitted linear regression line are generated for all three languages in cells 84, 85, and 86.

Code Feedback:

Concept Explanation & Output Analysis: The plots correctly visualize the linear relationship in log-log space. Each plot shows the empirical data as a scatter plot and the model as a straight line, which is the correct way to represent a linear regression fit for Heaps' Law.

Item 15 (Cell 92)

Rubric Criteria: T5.1. Zipf's Summary Table

Score: 5.0/5.0 - Full

Justification: Output: N/A (Markdown Cell). Cell 92 contains a single markdown table that compares Zipf and Mandelbrot parameters (including) and fit metrics across all three languages. The accompanying text interpretation identifies Mandelbrot as the better-fitting model based on these metrics.

Code Feedback:

Concept Explanation & Output Analysis: A comprehensive summary table effectively compares the key parameters and fit metrics for the Zipf/Mandelbrot analysis across all three languages. This allows for easy cross-linguistic comparison and supports the subsequent discussion.

Item 16 (Cell 99)

Rubric Criteria: T5.2. Heaps' Summary Table

Score: 5.0/5.0 - Full

Justification: Output: N/A (Markdown Cell). Cell 99 contains a markdown table titled 'Heaps' Law Parameters and Corpus Statistics' that compares the K and values (from curve fitting) for English, Marathi, and Hindi.

Code Feedback:

Concept Explanation & Output Analysis: A clear summary table is provided that compares the Heaps' Law parameters K and across the three languages, facilitating the comparative analysis presented in the report.

Item 17 (Cell 109)

Rubric Criteria: T5.3. Memory Summary Table

Score: 5.0/5.0 - Full

Justification: Output: N/A (Markdown Cell). Cell 109 provides a markdown table titled 'Vocabulary–Document Matrix Memory Estimates' that summarizes and compares the final calculated memory requirements for all three languages.

Code Feedback:

Concept Explanation & Output Analysis: The submission includes a final summary table that consolidates the memory calculation results for all three languages, clearly presenting the final estimated memory sizes side-by-side.

Item 18 (Cell 32)

Rubric Criteria: T5.4. Discussion Formatting

Score: 5.0/5.0 - Full

Justification: Output: N/A (Markdown Cell). The notebook contains numerous, well-formatted discussion sections (e.g., cells 32, 43, 55, 66, 88, 92, etc.) that use markdown features such as headers, bold text, and bullet points to structure the interpretation of results.

Code Feedback:

Concept Explanation & Output Analysis: The student makes extensive use of markdown to format all textual analysis and discussion. This includes headers, lists, bolding, and tables, which greatly improves the readability and professional presentation of the report.
